** Basic C Programming Questions **

** Character Programs **

Q80. Write a C program to check whether a character is vowel or consonant.
Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   char ch;
   clrscr();
   printf("Enter a character: ");
   scanf("%c", &ch);
   if(ch == 'a' || ch == 'e' || ch == 'i' ||
     ch == 'o' || ch == 'u' ||
     ch == 'A' || ch == 'E' || ch == 'I' ||
     ch == 'O' || ch == 'U')
      printf("Vowel");
   else
      printf("Consonant");
   getch();
}
```

Output: -
```
Enter a character: A
 Vowel
```

Q81. Write a C program to check whether a character is alphabet, digit or special character.
Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   char ch;
   clrscr();
   printf("Enter a character: ");
   scanf("%c", &ch);
   if((ch >= 'A' && ch <= 'Z') ||
     (ch >= 'a' && ch <= 'z'))
      printf("Alphabet");
   else if(ch >= '0' && ch <= '9')
      printf("Digit");
   else
      printf("Special Character");
   getch();
}
```

Output: -
```
Enter a character: 7
 Digit
```

** Switch Statement **

Q82. Write a C program to perform arithmetic operations using switch case.
Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int a, b, choice;
   clrscr();
   printf("Enter two numbers: ");
```

```
    scanf("%d%d", &a, &b);
    printf("\n1. Addition");
    printf("\n2. Subtraction");
    printf("\n3. Multiplication");
    printf("\n4. Division");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
      case 1:
        printf("Addition = %d", a + b);
        break;
      case 2:
        printf("Subtraction = %d", a - b);
        break;
      case 3:
        printf("Multiplication = %d", a * b);
        break;
      case 4:
        printf("Division = %d", a / b);
        break;
      default:
        printf("Invalid Choice");
    }
    getch();
}
```

Output: - Enter two numbers: 20 5

```
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 1
Addition = 25
```

Matrix

Q83. Write a C program to input and display a 2×2 matrix.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
  int a[2][2], i, j;
  clrscr();
  printf("Enter matrix elements:\n");
  for(i = 0; i < 2; i++)
  {
    for(j = 0; j < 2; j++)
      scanf("%d", &a[i][j]);
  }
  printf("Matrix:\n");
  for(i = 0; i < 2; i++)
  {
    for(j = 0; j < 2; j++)
      printf("%d ", a[i][j]);
    printf("\n");
  }
  getch();
```

```
}
```

Output: - Enter matrix elements:
1 2
3 4

Matrix:
1 2
3 4

Q84. Write a C program to add two 2×2 matrices.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int a[2][2], b[2][2], c[2][2];
   int i, j;
   clrscr();
   printf("Enter first matrix:\n");
   for(i = 0; i < 2; i++)
      for(j = 0; j < 2; j++)
         scanf("%d", &a[i][j]);
   printf("Enter second matrix:\n");
   for(i = 0; i < 2; i++)
      for(j = 0; j < 2; j++)
         scanf("%d", &b[i][j]);
   for(i = 0; i < 2; i++)
   {
      for(j = 0; j < 2; j++)
         c[i][j] = a[i][j] + b[i][j];
   }
   printf("Sum Matrix:\n");
   for(i = 0; i < 2; i++)
   {
      for(j = 0; j < 2; j++)
         printf("%d ", c[i][j]);
      printf("\n");
   }
   getch();
}
```

Output: - Enter first matrix:
1 2
3 4

Enter second matrix:
5 6
7 8

Sum Matrix:
6 8
10 12

Q85. Write a C program to find the sum of all even digits of a number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n, d, sum = 0;
```

```
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &n);
    while(n > 0)
    {
      d = n % 10;
      if(d % 2 == 0)
        sum = sum + d;
      n = n / 10;
    }
    printf("Sum of Even Digits = %d", sum);
    getch();
}
```

Output: - Enter a number: 123456
Sum of Even Digits = 12

Q86. Write a C program to find the sum of all odd digits of a given number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int n, digit, sum = 0;
  clrscr();
  printf("Enter a number: ");
  scanf("%d", &n);
  while(n > 0)
  {
    digit = n % 10;
    if(digit % 2 != 0)
    {
      sum = sum + digit;
    }
    n = n / 10;
  }
  printf("Sum of odd digits = %d", sum);
  getch();
}
```

Output: - Enter a number: 123456
Sum of odd digits = 9

Q87. Write a C program to count the number of even digits in a given number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int n, digit, count = 0;
  clrscr();
  printf("Enter a number: ");
  scanf("%d", &n);
  while(n > 0)
  {
    digit = n % 10;
    if(digit % 2 == 0)
    {
      count++;
    }
    n = n / 10;
```

```
    }
    printf("Number of even digits = %d", count);
    getch();
}
```

Output: - Enter a number: 123456
Number of even digits = 3

Q88. Write a C program to count the number of odd digits in a given number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n, digit, count = 0;
   clrscr();
   printf("Enter a number: ");
   scanf("%d", &n);
   while(n > 0)
   {
      digit = n % 10;
      if(digit % 2 != 0)
      {
         count++;
      }
      n = n / 10;
   }
   printf("Number of odd digits = %d", count);
   getch();
}
```

Output: - Enter a number: 123456
Number of odd digits = 3

Q89. Write a C program to find the first digit of a given number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n, first;
   clrscr();
   printf("Enter a number: ");
   scanf("%d", &n);
   while(n >= 10)
   {
      n = n / 10;
   }
   first = n;
   printf("First digit = %d", first);
   getch();
}
```

Output: - Enter a number: 56789
First digit = 5

Q90. Write a C program to find the last digit of a number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
```

```
    int n, last;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &n);
    last = n % 10;
    printf("Last digit = %d", last);
    getch();
}
```

Output: - Enter a number: 56789
Last digit = 9

Q91. Write a C program to find the sum of first and last digit of a number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
    int n, first, last, sum;
    clrscr();
    printf("Enter a number: ");
    scanf("%d", &n);
    last = n % 10;
    while(n >= 10)
    {
       n = n / 10;
    }
    first = n;
    sum = first + last;
    printf("First digit = %d", first);
    printf("\nLast digit = %d", last);
    printf("\nSum = %d", sum);
    getch();
}
```

Output: - Enter a number: 56789
First digit = 5
Last digit = 9
Sum = 14

Q92. Write a C program to find the middle digit of a three-digit number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
    int n, middle;
    clrscr();
    printf("Enter a three-digit number: ");
    scanf("%d", &n);
    middle = (n / 10) % 10;
    printf("Middle digit = %d", middle);
    getch();
}
```

Output: - Enter a three-digit number: 456
Middle digit = 5

Q93. Write a C program to find the reverse of a number and its sum with the original number.

Ans: -
```
#include <stdio.h>
#include <conio.h>
```

```
void main()
{
   int n, original, digit, reverse = 0, sum;
   clrscr();
   printf("Enter a number: ");
   scanf("%d", &n);
   original = n;
   while(n > 0)
   {
      digit = n % 10;
      reverse = reverse * 10 + digit;
      n = n / 10;
   }
   sum = original + reverse;
   printf("Original number = %d", original);
   printf("\nReverse number = %d", reverse);
   printf("\nSum = %d", sum);
   getch();
}
```

Output: - Enter a number: 123
Original number = 123
Reverse number = 321
Sum = 444

Q94. Write a C program to find the reverse of a number and product of its digits.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n, digit, reverse = 0, product = 1;
   clrscr();
   printf("Enter a number: ");
   scanf("%d", &n);
   while(n > 0)
   {
      digit = n % 10;
      reverse = reverse * 10 + digit;
      product = product * digit;
      n = n / 10;
   }
   printf("Reverse = %d", reverse);
   printf("\nProduct of digits = %d", product);
   getch();
}
```

Output: - Enter a number: 123
Reverse = 321
Product of digits = 6

Q95. Write a C program to check whether a number is divisible by 2, 3 and 5.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n;
   clrscr();
   printf("Enter a number: ");
   scanf("%d", &n);
```

```
    if(n % 2 == 0 && n % 3 == 0 && n % 5 == 0)
    {
      printf("Number is divisible by 2, 3 and 5");
    }
    else
    {
      printf("Number is not divisible by 2, 3 and 5");
    }
    getch();
  }
```

Output: - Enter a number: 60
Number is divisible by 2, 3 and 5

Q96. Write a C program to check whether a number is divisible by 7.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int n;
  clrscr();
  printf("Enter a number: ");
  scanf("%d", &n);
  if(n % 7 == 0)
  {
    printf("Number is divisible by 7");
  }
  else
  {
    printf("Number is not divisible by 7");
  }
  getch();
}
```

Output: - Enter a number: 49
Number is divisible by 7

Q97. Write a C program to check whether a number is divisible by 9.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int n;
  clrscr();
  printf("Enter a number: ");
  scanf("%d", &n);
  if(n % 9 == 0)
  {
    printf("Number is divisible by 9");
  }
  else
  {
    printf("Number is not divisible by 9");
  }
  getch();
}
```

Output: - Enter a number: 81
Number is divisible by 9

Q98. Write a C program to convert a decimal number into binary.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n, remainder, binary = 0, place = 1;
   clrscr();
   printf("Enter a decimal number: ");
   scanf("%d", &n);
   while(n > 0)
   {
      remainder = n % 2;
      binary = binary + remainder * place;
      place = place * 10;
      n = n / 2;
   }
   printf("Binary number = %d", binary);
   getch();
}
```

Output: - Enter a decimal number: 10
Binary number = 1010

Q99. Write a C program to convert a binary number into decimal.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int binary, digit, decimal = 0, place = 1;
   clrscr();
   printf("Enter a binary number: ");
   scanf("%d", &binary);
   while(binary > 0)
   {
      digit = binary % 10;
      decimal = decimal + digit * place;
      place = place * 2;
      binary = binary / 10;
   }
   printf("Decimal number = %d", decimal);
   getch();
}
```

Output: - Enter a binary number: 1010
Decimal number = 10

Q100. Write a C program to convert a decimal number into octal.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int n, remainder, octal = 0, place = 1;
   clrscr();
   printf("Enter a decimal number: ");
   scanf("%d", &n);
   while(n > 0)
   {
      remainder = n % 8;
      octal = octal + remainder * place;
```

```
        place = place * 10;
        n = n / 8;
      }
      printf("Octal number = %d", octal);
      getch();
    }
```

Output: - Enter a decimal number: 25
Octal number = 31

Q101. Write a C program to convert a decimal number into hexadecimal.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
  int n, remainder;
  char hex[20];
  int i = 0;
  clrscr();
  printf("Enter a decimal number: ");
  scanf("%d", &n);
  if(n == 0)
  {
    printf("Hexadecimal = 0");
  }
  else
  {
    while(n > 0)
    {
      remainder = n % 16;
      if(remainder < 10)
        hex[i] = remainder + '0';
      else
        hex[i] = remainder - 10 + 'A';
      i++;
      n = n / 16;
    }
    printf("Hexadecimal = ");
    while(i > 0)
    {
      i--;
      printf("%c", hex[i]);
    }
  }
  getch();
}
```

Output: - Enter a decimal number: 255
Hexadecimal = FF

Q102. Write a C program to convert a binary number into octal.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
  int binary, digit;
  int decimal = 0, place = 1;
  int octal = 0, octalPlace = 1;
```

```
    clrscr();
    printf("Enter a binary number: ");
    scanf("%d", &binary);
    while(binary > 0)
    {
      digit = binary % 10;
      decimal = decimal + digit * place;
      place = place * 2;
      binary = binary / 10;
    }
    while(decimal > 0)
    {
      digit = decimal % 8;
      octal = octal + digit * octalPlace;
      octalPlace = octalPlace * 10;
      decimal = decimal / 8;
    }
    printf("Octal number = %d", octal);
    getch();
  }
```

Output: - Enter a binary number: 1010
Octal number = 12

Q103. Write a C program to convert an octal number into decimal.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int octal, digit;
  int decimal = 0, place = 1;
  clrscr();
  printf("Enter an octal number: ");
  scanf("%d", &octal);
  while(octal > 0)
  {
    digit = octal % 10;
    decimal = decimal + digit * place;
    place = place * 8;
    octal = octal / 10;
  }
  printf("Decimal number = %d", decimal);
  getch();
}
```

Output: - Enter an octal number: 31
Decimal number = 25

Q104. Write a C program to check whether a number is a palindrome.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int n, original, digit, reverse = 0;
  clrscr();
  printf("Enter a number: ");
  scanf("%d", &n);
  original = n;
  while(n > 0)
```

```
    {
      digit = n % 10;
      reverse = reverse * 10 + digit;
      n = n / 10;
    }
    if(original == reverse)
      printf("Number is Palindrome");
    else
      printf("Number is not Palindrome");
    getch();
  }
```

Output: - Enter a number: 121
Number is Palindrome

Q105. Write a C program to print a right-aligned star triangle.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int i, j, n;
  clrscr();
  printf("Enter number of rows: ");
  scanf("%d", &n);
  for(i = 1; i <= n; i++)
  {
    for(j = 1; j <= n - i; j++)
    {
      printf("  ");
    }
    for(j = 1; j <= i; j++)
    {
      printf("* ");
    }
    printf("\n");
  }
  getch();
}
```

Output: - Enter number of rows: 5

```
        *
      * *
    * * *
  * * * *
* * * * *
```

Q106. Write a C program to print an inverted right-aligned star triangle.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
  int i, j, n;
  clrscr();
  printf("Enter number of rows: ");
  scanf("%d", &n);
  for(i = n; i >= 1; i--)
  {
    for(j = 1; j <= n - i; j++)
```

```
        {
          printf("  ");
        }
        for(j = 1; j <= i; j++)
        {
          printf("* ");
        }
        printf("\n");
      }
      getch();
    }
```

Output: - Enter number of rows: 5

```
    * * * * *
     * * * *
      * * *
       * *
        *
```

Q107. Write a C program to print a star pyramid.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
  int i, j, n;
  clrscr();
  printf("Enter number of rows: ");
  scanf("%d", &n);
  for(i = 1; i <= n; i++)
  {
    for(j = 1; j <= n - i; j++)
    {
      printf(" ");
    }
    for(j = 1; j <= 2 * i - 1; j++)
    {
      printf("*");
    }
    printf("\n");
  }
  getch();
}
```

Output: - Enter number of rows: 5

```
      *
     ***
    *****
   *******
  *********
```

Q108. Write a C program to print an inverted star pyramid.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
  int i, j, n;
  clrscr();
```

```
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = n; i >= 1; i--)
    {
      for(j = 1; j <= n - i; j++)
      {
        printf(" ");
      }
      for(j = 1; j <= 2 * i - 1; j++)
      {
        printf("*");
      }
      printf("\n");
    }
    getch();
  }
```

Output: - Enter number of rows: 5

```
    *********
     *******
      *****
       ***
        *
```

Q109. Write a C program to print a diamond star pattern.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
  int i, j, n;
  clrscr();
  printf("Enter number of rows: ");
  scanf("%d", &n);
  for(i = 1; i <= n; i++)
  {
    for(j = 1; j <= n - i; j++)
      printf(" ");
    for(j = 1; j <= 2 * i - 1; j++)
      printf("*");
    printf("\n");
  }
  for(i = n - 1; i >= 1; i--)
  {
    for(j = 1; j <= n - i; j++)
      printf(" ");
    for(j = 1; j <= 2 * i - 1; j++)
      printf("*");
    printf("\n");
  }
  getch();
}
```

Output: - Enter number of rows: 5

```
      *
     ***
    *****
   *******
```

```
*********
 *******
  *****
   ***
    *
```

Q110. Write a C program to print a hollow square star pattern.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter size of square: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= n; j++)
      {
         if(i == 1 || i == n || j == 1 || j == n)
            printf("* ");
         else
            printf("  ");
      }
      printf("\n");
   }
   getch();
}
```

Output: - Enter size of square: 5

```
* * * * *
*       *
*       *
*       *
* * * * *
```

Q111. Write a C program to print a hollow rectangle.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, rows, cols;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &rows);
   printf("Enter number of columns: ");
   scanf("%d", &cols);
   for(i = 1; i <= rows; i++)
   {
      for(j = 1; j <= cols; j++)
      {
         if(i == 1 || i == rows || j == 1 || j == cols)
            printf("* ");
         else
            printf("  ");
      }
      printf("\n");
   }
```

```
    getch();
}
```

Output: - Enter number of rows: 4
Enter number of columns: 7

```
* * * * * * *
*           *
*           *
* * * * * * *
```

Q112. Write a C program to print a hollow right triangle.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= i; j++)
      {
         if(j == 1 || j == i || i == n)
            printf("* ");
         else
            printf("  ");
      }
      printf("\n");
   }
   getch();
}
```

Output: - Enter number of rows: 5

```
*
* *
*   *
*     *
* * * * *
```

Q113. Write a C program to print a hollow pyramid.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= n - i; j++)
         printf(" ");
      for(j = 1; j <= 2 * i - 1; j++)
      {
         if(j == 1 || j == 2 * i - 1 || i == n)
            printf("*");
```

```
            else
               printf(" ");
         }
         printf("\n");
      }
      getch();
   }
```

Output: - Enter number of rows: 5

```
     *
    * *
   *   *
  *     *
 *********
```

Q114. Write a C program to print a hollow diamond.

Ans: -

```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= n - i; j++)
         printf(" ");
      for(j = 1; j <= 2 * i - 1; j++)
      {
         if(j == 1 || j == 2 * i - 1)
            printf("*");
         else
            printf(" ");
      }
      printf("\n");
   }
   for(i = n - 1; i >= 1; i--)
   {
      for(j = 1; j <= n - i; j++)
         printf(" ");
      for(j = 1; j <= 2 * i - 1; j++)
      {
         if(j == 1 || j == 2 * i - 1)
            printf("*");
         else
            printf(" ");
      }
      printf("\n");
   }
   getch();
}
```

Output: - Enter number of rows: 5

```
     *
    * *
   *   *
```

```
 *   *
*    *
 *   *
  * *
   **
    *
```

Q115. Write a C program to print a number pyramid.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= n - i; j++)
         printf(" ");
      for(j = 1; j <= i; j++)
         printf("%d ", j);
      printf("\n");
   }
   getch();
}
```

Output: - Enter number of rows: 5

```
    1
   1 2
  1 2 3
 1 2 3 4
1 2 3 4 5
```

Q116. Write a C program to print the same number in each row.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= i; j++)
      {
         printf("%d ", i);
      }
      printf("\n");
   }
   getch();
}
```

Output: - Enter number of rows: 5

```
1
2 2
3 3 3
```

```
4 4 4 4
5 5 5 5 5
```

Q117. Write a C program to print a reverse number triangle.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = n; i >= 1; i--)
   {
      for(j = 1; j <= i; j++)
      {
         printf("%d ", j);
      }
      printf("\n");
   }
   getch();
}
```

Output: - Enter number of rows: 5
```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Q118. Write a C program to print a reverse number pyramid.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void main()
{
   int i, j, n;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = n; i >= 1; i--)
   {
      for(j = 1; j <= n - i; j++)
         printf(" ");
      for(j = 1; j <= i; j++)
         printf("%d ", j);
      printf("\n");
   }
   getch();
}
```

Output: - Enter number of rows: 5
```
1 2 3 4 5
 1 2 3 4
  1 2 3
   1 2
    1
```

** Function **

Q119. Write a C program to input two numbers and find their sum using a function with arguments.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void sum(int a, int b)
{
   int result;
   result = a + b;
   printf("Sum = %d", result);
}
void main()
{
   int x, y;
   clrscr();
   printf("Enter first number: ");
   scanf("%d", &x);
   printf("Enter second number: ");
   scanf("%d", &y);
   sum(x, y);
   getch();
}
```

Output: -
```
Enter first number: 25
Enter second number: 15
Sum= 40
```

Q120. Write a C program to input two numbers and find their multiplication using a function with arguments and return value.

Ans: -
```
#include <stdio.h>
#include <conio.h>
int multiply(int a, int b)
{
   int result;
   result = a * b;
   return result;
}
void main()
{
   int x, y, ans;
   clrscr();
   printf("Enter first number: ");
   scanf("%d", &x);
   printf("Enter second number: ");
   scanf("%d", &y);
   ans = multiply(x, y);
   printf("Multiplication = %d", ans);
   getch();
}
```

Output: -
```
Enter first number: 8
Enter second number: 7
Multiplication = 56
```

Q121. Write a C program to input a number and display its square using a function without arguments and without return value.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void square()
```

```
{
   int n, result;
   printf("Enter a number: ");
   scanf("%d", &n);
   result = n * n;
   printf("Square = %d", result);
}
void main()
{
   clrscr();
   square();
   getch();
}
```

Output: - Enter a number: 12
Square = 144

Q122. Write a C program to input three numbers and find the largest number using a function with arguments and return value.

Ans: -
```
#include <stdio.h>
#include <conio.h>
int largest(int a, int b, int c)
{
   int max;
   max = a;
   if (b > max)
      max = b;
   if (c > max)
      max = c;
   return max;
}
void main()
{
   int x, y, z, ans;
   clrscr();
   printf("Enter first number: ");
   scanf("%d", &x);
   printf("Enter second number: ");
   scanf("%d", &y);
   printf("Enter third number: ");
   scanf("%d", &z);
   ans = largest(x, y, z);
   printf("Largest = %d", ans);
   getch();
}
```

Output: - Enter first number: 25
Enter second number: 48
Enter third number: 32
Largest = 48

Q123. Write a C program to swap two numbers using Call by Value.

Ans: -
```
#include <stdio.h>
#include <conio.h>
void swap(int a, int b)
{
   int temp;
   temp = a;
   a = b;
   b = temp;
   printf("\nInside Function:");
   printf("\na = %d, b = %d", a, b);
}
void main()
{
   int x, y;
   clrscr();
   printf("Enter two numbers: ");
   scanf("%d%d", &x, &y);
   swap(x, y);
   printf("\nOutside Function:");
   printf("\nx = %d, y = %d", x, y);
   getch();
}
```

Output: -
```
Enter two numbers: 10 20
 Inside Function:
 a = 20, b = 10
 Outside Function:
 x = 10, y = 20
```

Q124. Write a C program to swap two numbers using pointers (Call by Reference).

Ans: -
```
#include <stdio.h>
#include <conio.h>
void swap(int *a, int *b)
{
   int temp;
   temp = *a;
   *a = *b;
   *b = temp;
}
void main()
{
   int x, y;
   clrscr();
   printf("Enter two numbers: ");
   scanf("%d%d", &x, &y);
   swap(&x, &y);
   printf("After Swapping:");
   printf("\nx = %d, y = %d", x, y);
   getch();
}
```

Output: -
```
Enter two numbers: 10 20
   After Swapping:
   x = 20, y = 10
```

Q125. Write a C program to find the factorial of a number using recursion.

Ans: -

```
#include <stdio.h>
#include <conio.h>
int fact(int n)
{
   if (n == 0 || n == 1)
      return 1;
   return n * fact(n - 1);
}
void main()
{
   int n, ans;
   clrscr();
   printf("Enter a number: ");
   scanf("%d", &n);
   if (n < 0)
      printf("Factorial is not defined for negative numbers");
   else
   {
      ans = fact(n);
      printf("Factorial = %d", ans);
   }
   getch();
}
```

Output: - Enter a number: 5
Factorial = 120

THANK YOU

NEXT

QUESTION

NEXT

PAGE